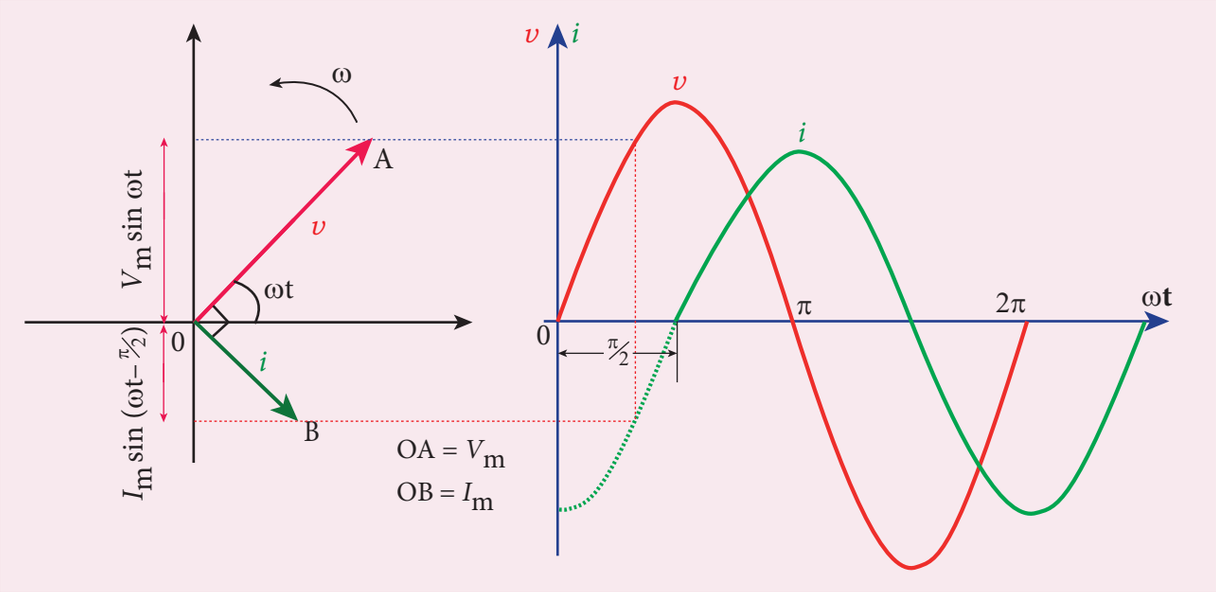


Figure 4.43 Phasor diagram and wave diagram for AC circuit with L

$$\left(\because \quad -\cos\omega t = -\sin\left(\pi/2 - \omega t\right) \right.$$
$$\left. = \sin\left(\omega t - \pi/2\right) \right)$$

$$i = \frac{V_m}{\omega L}\sin\left(\omega t - \pi/2\right)$$
$$\text{or} \quad i = I_m \sin\left(\omega t - \pi/2\right) \qquad (4.41)$$

where $\frac{V_m}{\omega L} = I_m$, the peak value of the alternating current in the circuit. From equation (4.40) and (4.41), it is evident that current lags behind the applied voltage by $\pi/2$ in an inductive circuit. This fact is depicted in the phasor diagram. In the wave diagram also, it is seen that current lags the voltage by 90° (Figure 4.43).

Inductive reactance X_L

The peak value of current I_m is given by $I_m = \frac{V_m}{\omega L}$. Let us compare this equation with $I_m = \frac{V_m}{R}$ from resistive circuit. The quantity ωL plays the same role as the resistance in resistive circuit. This is the resistance offered by the inductor, called inductive reactance (X_L). It is measured in ohm.

$$X_L = \omega L$$

An inductor blocks AC but it allows DC. Why? and How?

An inductor L is a closely wound helical coil. The steady DC current flowing through L produces uniform magnetic field around it and the magnetic flux linked remains constant. Therefore there is no self-induction and self-induced emf (back emf). Since inductor behaves like a resistor, DC flows through an inductor.

The AC flowing through L produces time-varying magnetic field which in turn induces self-induced emf (back emf). This back emf, according to Lenz's law, opposes any change in the current. Since AC varies both in magnitude and direction, its flow is opposed in L. For an ideal inductor of zero ohmic resistance, the back emf is equal and opposite to the applied emf. Therefore L blocks AC.

12th Physics_Vol-1_Unit 4_English Medium.indd 240 23-12-2022 17:58:18